CASE STUDY REPORT

A CASE STUDY ON ATM WITHDRAWAL CONSUMPTION TREND, REFILL OPTIMIZATION & CUSTOMER SERVICE ENHANCEMENT.

A report submitted in partial fulfilment of the requirements for the award of a degree in

MASTER OF SCIENCE

(FINANCIAL TECHNOLOGY)

by

ANCILLA FIONA DSOUZA

RA2532271010013

UNDER THE GUIDANCE OF

Sr. Edu. Johnsundar J
Trainer – Data Analytics and FinTech

DEPARTMENT OF ECONOMICS

SRM INSTITUTE OF SCIENCE AND Humanities,

Kattankulatbur, Chennai.

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CERTIFICATE

This is to clarify that the Data Analytics and Fintech Project Report entitled on "ATM Withdrawal consumption, refill optimization & Customer serve Enhancement" has been carried out by Ancilla Fiona Dsouza, (Registration number: RA2532271010013) A student of MS FinTech, Department of Economics, Faculty of Science and Humanities, SRM Institute of Science and Technology, Kattankulathur, under my guidance.

This project work is being submitted in partial fulfilment of the requirement for the award of degree of Master of Science in Financial Technology. The work embodied in the report is original and has not been submitted earlier for the award of any degree or diploma of this or any other university.

PROJECT GUIDE
Mr. Johnsundar J
Trainer (Imarticus)
Financial management
SRM Institute of Science and Technology
Kattankulathur

THE HEAD OF DEPARTMENT
Mr. Prakash V
Head and Associate Professor
Department of Economics
SRM Institute of Science and Technology
Kattankulathur

Examiner(s):

1. ___________________

2. ___________________

Approved by

External Examiner

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I am also thankful for the support provided by my college and the access to essential software and tools that enabled me to carry out the data analysis work. The use of Python and Power BI greatly helped me in cleaning, processing, and visualizing the data required for this project.

DECLARATION

I **Ancilla Dsouza**, Registered number: **RA2532271010013**, hereby declare that this Project report done is the original work done by me and submitted to the **DEPARTMENT OF ECONOMICS, FACULTY OF SCIENCE AND HUMANITIES,** in the partial fulfilment of requirements for the award of the degree of **M.S. FINTECH** under the guidance of **Sr. Johnsundar J, Trainer in Data Analytics and FinTech.**

Date:15-12-2025 **Ancilla Fiona Dsouza**

Place: Chennai **RA2332201010013**

ABSTRACT

This project focuses on analyzing **ATM withdrawal consumption trends and optimizing refill requirements and customers service enhancement** using data analytics. The dataset was collected from an online source and processed using **Excel and Python** for cleaning, transformation, and exploratory data analysis. Machine learning models such as **classification and regression** were applied to predict refill needs and cash withdrawal amounts. A **Power BI** dashboard was created to visualize patterns, trends, and ATM performance insights. The study helps **improve cash availability**, reduce unnecessary refill trips, and enhance user convenience through data-driven decision-making.

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CHAPTER 1

INTRODUCTION

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About the Domain:

FinTech combines technology with financial services to improve efficiency in banking. Data Analytics helps organizations study data, identify trends, and make better decisions. In ATM operations, it supports understanding withdrawal patterns and planning cash refills effectively.

Background of the Study:

ATMs must always provide cash availability to customers. Poor cash forecasting leads to excess cash lying idle, loss of interest, and unnecessary refill trips. These refill trips involve high costs due to vans, custodians, and security staff. Women often feel unsafe using ATMs at night, and senior citizens struggle with small text and complex steps. Analyzing ATM data helps improve both operational efficiency and user experience.

Problem Statement:

Banks face issues like incorrect cash demand prediction, frequent refill trips, and higher operational costs. In addition, women face safety concerns at ATMs, and senior citizens struggle with ATM usage due to design difficulties. This project focuses on solving these challenges using data analytics

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Need for the Project:

There is a need to optimize ATM refills, reduce operational cost, ensure higher ATM uptime, and improve customer convenience. Data analysis helps predict cash usage, identify refill requirements, and understand user challenges more effectively.

Scope of the Project:

The project includes data collection, cleaning, and analysis using Excel and Python, applying classification and regression models, and building a Power BI dashboard. It focuses only on the provided dataset and analytical results

Objectives of the Project:

- To analyze ATM withdrawal trends.
- To predict refill requirements using classification.
- To estimate cash withdrawal amounts using regression.
- To create a Power BI dashboard for insights.
- To understand user difficulties, especially women and senior citizens.
- To suggest improvements for better ATM operations.

CHAPTER 2

THEORETICAL BACKGROUND

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Introduction:

Financial Data Analytics (FDA) is the use of data analysis techniques to support decision-making in financial operations. In banking, it helps predict trends, forecast demand, optimize processes, and improve customer service. This study focuses on applying FDA to ATM withdrawal data to enhance cash management and operational efficiency.

Role of Data in ATM Management:

Data analysis enables banks to:

- Understand ATM cash withdrawal patterns.
- Predict cash demand and refill needs.
- Reduce unnecessary Cash-in-Transit (CIT) trips.
- Improve ATM uptime and availability.
- Identify challenges faced by specific users, such as women and senior citizens.

Overview of the Selected Topic:

The project analyzes **ATM Withdrawal Consumption Trends** and develops strategies for **Refill Optimization**. It involves:

- Collecting ATM transaction data.
- Cleaning and transforming data using Excel and Python.
- Performing exploratory data analysis (EDA).
- Applying classification to predict refill needs.
- Applying regression to forecast cash withdrawals.
- Visualizing insights using a Power BI dashboard.

Tools & Technologies Used:

- **Microsoft Excel:** Raw data handling, creating new columns, removing duplicates.
- **Python (Google Colab):** Data cleaning, EDA, classification and regression models.
- **Power BI:** Interactive dashboard for visual analysis and insights.

Summary:

This chapter provides the theoretical foundation for using data analytics in ATM operations. Understanding cash usage patterns and predicting refill requirements are essential to improve efficiency, reduce cost, and enhance customer satisfaction.

CHAPTER 3
DATASET
DESCRIPTION

Source of Data

The dataset was collected from the **GoMask.ai platform**, containing ATM transaction records for multiple ATMs. Each row represents a single ATM on a particular date, detailing cash withdrawals and refill requirements.

Data Fields / Attributes

The main columns include:

- **ATM_ID:** Unique identifier for each ATM
- **Date:** Transaction date
- **Cash_Withdrawn:** Total cash withdrawn on that date
- **Refill_needed_flag:** Indicates if the ATM requires a refill (1 = Yes, 0 = No)
- **Additional columns** created during preprocessing to support analysis (e.g., day, month, withdrawal categories)

Data Limitations

- Limited number of rows (approx. 200+), which may not cover all ATMs in a city.
- No geographic information about ATM locations.
- Does not account for holidays, events, or ATM downtime beyond the dataset.

Sampl

ATM_ID	Branch	location_city	location_street_address	location_postal_code	location_country	Max_Cash_Capacity
ATM101	Main Street Branch	San Francisco	55 Main St	94105	USA	13040
ATM101	Downtown Branch	New York	123 Main St	10001	USA	13040
ATM102	Grand Plaza Mall	Los Angeles	456 Market Ave	90017	USA	15420
ATM103	Main St Branch	Houston	100 Main St	77002	USA	13000
ATM103	Eastside Grocery	Chicago	789 Broadway	60601	USA	13000
ATM104	Union Station	Boston	1 Metro Dr	02118	USA	18000
ATM105	Civic Center Mall	Los Angeles	223 S Main St	90012	USA	11980
ATM105	City Center Bank	Houston	56 Elm St	77002	USA	11980
ATM106	Lakeside Mall	Minneapolis	1500 Lake Dr	55402	USA	12000
ATM107	24Hr Pharmacy	Dallas	99 Madison Ave	75201	USA	8000
ATM108	Airport Concourse	Atlanta	1 Airport Rd	30320	USA	12500
ATM109	Eastside Bank	Tulsa	111 East 7th St	74119	USA	13070
ATM109	Central Park Kiosk	New York	2 Park Ave	10022	USA	13070
ATM110	Market Square	San Francisco	135 Oak St	94102	USA	11445
ATM116	Beacon Bank Hall	Boston	15 Beacon St	02108	USA	9640
ATM117	Harbor Bank	San Diego	200 Harbor Dr	92101	USA	9000
ATM118	Sunnyvale Plaza	Sunnyvale	784 Willow St	94086	USA	18000
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ATM120	Union Station	Washington	50 Massachusetts Ave NE	20002	USA	10000
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Sample of Raw ATM Transaction Data

Data Collection Method

ATM_ID	Branch	location_city	location_street_address	location_postal_code	location_country	Max_Cash_Capacity
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Cleaned ATM Dataset Ready for Analysis

CHAPTER 4
METHODOLOGY

Project Workflow

The project follows these steps:

- **Data Collection:** Downloaded ATM transaction dataset from GoMask.ai in Excel format.
- **Data Cleaning:** Removed duplicates, handled missing values, converted date formats, and created additional features in Excel and Python.
- **Exploratory Data Analysis (EDA):** Analyzed cash withdrawal trends, refill patterns, and generated visualizations in Python.
- **Machine Learning Models:**
  - **Classification:** Predict refill requirement (refill_needed_flag) using Random Forest Classifier.
  - **Regression:** Predict cash withdrawal amounts using Random Forest Regressor.
- **Results Export:** Exported classification and regression outputs to Excel.
- **Power BI Dashboard:** Connected cleaned dataset and model outputs to create an interactive dashboard with charts and filters for ATM cash usage insights.

Data Cleaning Process

- Converted date columns to proper datetime format.
- Checked and removed duplicate rows and repeated columns.
- Handled missing or inconsistent values.
- Created numeric and categorical features to support analysis (e.g., day, month, withdrawal category).
- Saved the cleaned dataset as **DAF_PROJECT_clean_for_PowerBI.xlsx** for further analysis.

Data Transformation

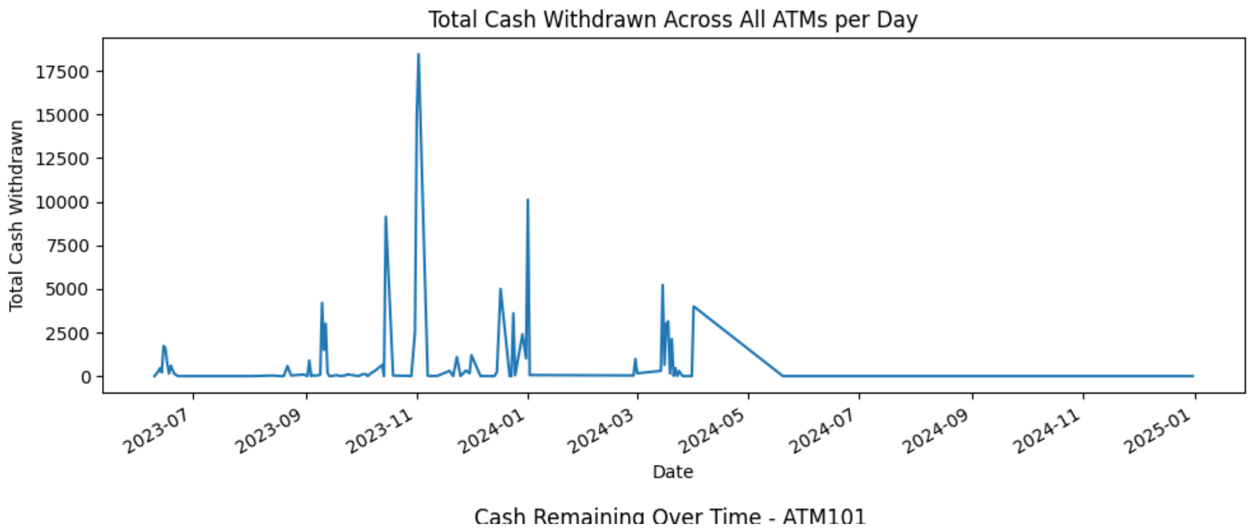
- Additional columns were created to aid analysis, such as day of week, month, and cash usage categories.
- Features were transformed into formats suitable for ML models.

Logic / Rules Applied (Non-ML)

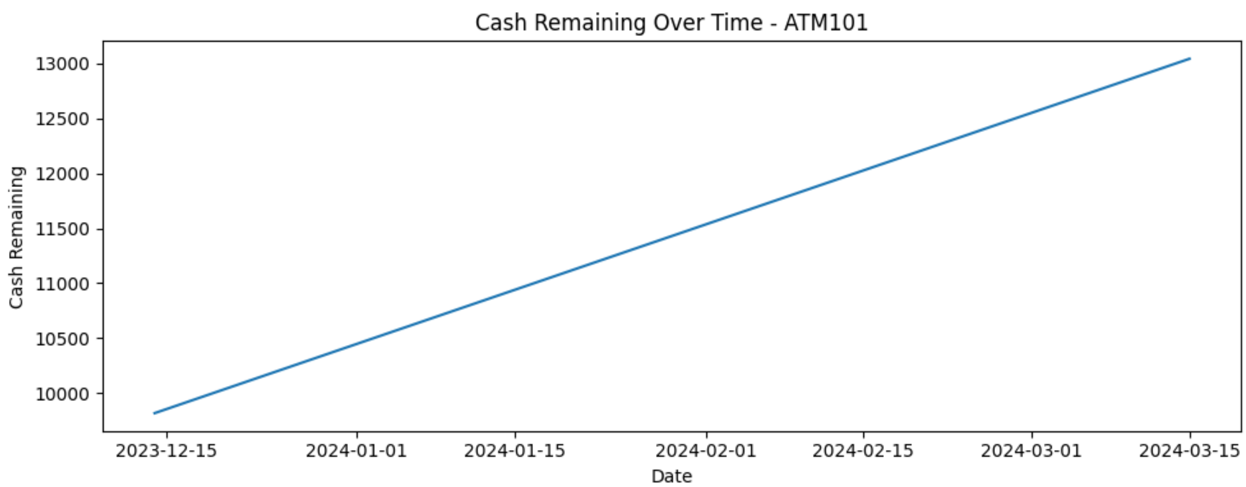
- Used Python and Excel to:
  - Remove duplicates
  - Handle missing data
  - Create derived columns for analysis
- Ensured dataset consistency before applying ML models.

Tools Implementation

- **Microsoft Excel:** Raw data checks, column modifications, basic cleaning.
- **Python (Google Colab):** Pandas for cleaning and transformation, Matplotlib/Seaborn for EDA, Scikit-learn for ML models.
- **Power BI:** Visualization of trends, patterns, and dashboard creation



Python – Data Cleaning & Output



Python – Trends in ATM Cash Usage

Summary

The methodology ensures accurate cleaning, transformation, and analysis of ATM withdrawal data. ML models predict refill requirements and cash demand, while Power BI provides actionable insights for ATM cash management.

CHAPTER 5

ANALYSIS & RESULTS

Exploratory Data Analysis (EDA)

After cleaning the dataset, EDA was performed to understand ATM cash withdrawal behavior.

- Analyzed trends in daily cash withdrawals.
- Examined patterns in refill requirements (refill_needed_flag).
- Identified ATMs with frequent refills or unusually high/low cash withdrawals.

Visualizations

- Line charts and bar charts were generated in Python to show trends in cash withdrawals over time.
- Refill requirement patterns were visualized to identify periods with high refill demand.
- Visualizations were also recreated in Power BI for interactive insights.

Classification & Regression Results

Classification Model (Refill Prediction):

- Algorithm: Random Forest Classifier
- Target: refill_needed_flag (0 = No, 1 = Yes)
- Performance: Accuracy = **92.5%**, Precision = 0.83, Recall = 0.71

Regression Model (Cash Withdrawal Prediction):

- Algorithm: Random Forest Regressor
- Target: Cash_Withdrawn
- Performance: MAE = 95.886, RMSE = 309.952, R^2 = 0.8292

	ATM_ID	Date	refill_needed_flag	predicted_refill_flag
9	ATM107	2024-03-18	0	0.0
18	ATM120	2023-10-13	0	0.0
55	ATM220	2023-06-18	0	0.0
75	ATM287	2023-12-29	0	0.0
150	ATM409	2024-03-22	0	0.0
104	ATM315	2023-06-15	0	0.0
135	ATM401	2024-07-15	0	0.0
137	ATM402	2023-11-01	1	0.0
164	ATM422	2023-06-12	0	0.0
76	ATM289	2023-06-21	0	0.0

Random Forest Classification Results for Refill Prediction

	ATM_ID	Date	Cash_Withdrawn	predicted_Cash_Withdrawn
9	ATM107	2024-03-18	30.0	-224.136543
18	ATM120	2023-10-13	160.0	-96.680455
55	ATM220	2023-06-18	0.0	-143.447603
75	ATM287	2023-12-29	2400.0	427.776807
150	ATM409	2024-03-22	380.0	1004.679794
104	ATM315	2023-06-15	0.0	468.632578
135	ATM401	2024-07-15	0.0	864.112274
137	ATM402	2023-11-01	3500.0	1428.315756
164	ATM422	2023-06-12	0.0	1047.672315
76	ATM289	2023-06-21	120.0	226.831833

```
Saved ALL_ATM_regression_results.xlsx
```

Random Forest Regression Results for Cash Withdrawal Prediction

Key Findings & Insights

- Cash withdrawal varies by day and ATM location.
- Refill is often required for ATMs with high daily withdrawals.
- ML models accurately predict refill needs and cash demand.
- Insights can help banks reduce unnecessary refill trips and improve ATM uptime.

Interpretation of Results

- High withdrawal periods indicate where more cash should be maintained.
- Refill prediction helps reduce operational costs.
- Patterns in data provide actionable information for planning ATM schedules.
- Recommendations for women and senior citizens’ convenience can be implemented based on user behavior insights.

CHAPTER 6

POWER BI DASHBOARD

Dashboard Overview

A Power BI dashboard was created using the cleaned dataset and outputs from the classification and regression models.
The dashboard provides interactive visualizations to monitor ATM cash withdrawal trends, refill requirements, and ATM performance insights.

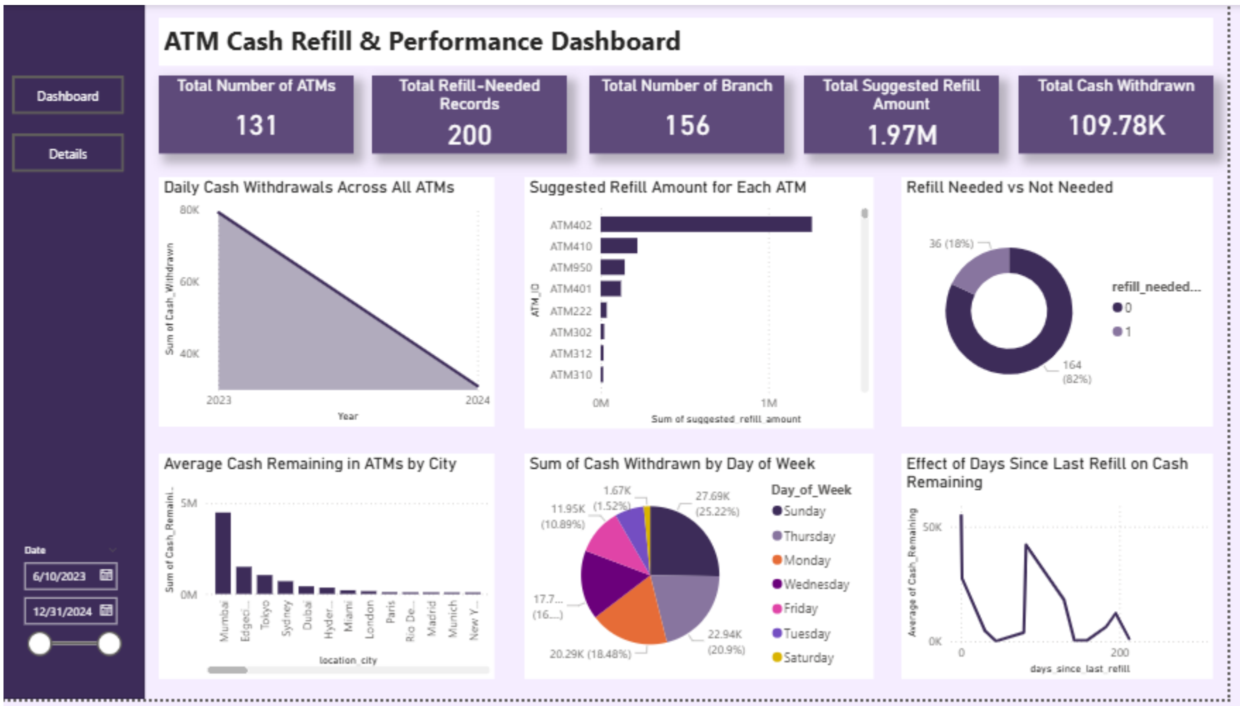
Visuals Used and Purpose

- **Line Charts:** Show daily cash withdrawal trends across ATMs.
- **Bar Charts:** Compare refill requirements among different ATMs.
- **Slicers / Filters:** Allow selection by ATM_ID and Date for detailed analysis.
- **Cards & KPI Visuals:** Display key metrics like total cash withdrawn and number of ATMs requiring refill.

Dashboard Insights

- Identifies ATMs with high withdrawal frequency and refill needs.
- Helps plan optimal refill schedules, reducing unnecessary CIT trips.
- Highlights periods with peak cash demand to maintain ATM uptime.
- Supports data-driven decisions to improve ATM efficiency and user satisfaction.

Screenshots



Power BI Dashboard Showing ATM Cash Withdrawal and Refill Insights

Summary

The Power BI dashboard consolidates cleaned data and model outputs to provide actionable insights. It allows bank managers to monitor ATM operations, optimize cash refills, and improve service availability effectively.

CHAPTER 7

CONCLUSION &

FUTURE ENHANCEMENT

Conclusion

The project successfully analyzed ATM cash withdrawal patterns and optimized refill planning using data analytics. Python and Excel were used for data cleaning and transformation, while classification and regression models accurately predicted refill requirements and cash demand. The Power BI dashboard provided interactive visual insights, helping to identify ATMs with high withdrawal frequency and improve operational efficiency. Overall, the study demonstrates how data-driven decisions can reduce unnecessary refill trips, maintain ATM uptime, and enhance user convenience.

Business Applications

- Reduce operational costs by optimizing Cash-in-Transit (CIT) trips.
- Maintain sufficient cash in ATMs to avoid downtime.
- Provide actionable insights for ATM managers to plan refills efficiently.
- Improve customer experience by addressing women's safety and senior citizen convenience.

Limitations

- The dataset is limited to 200+ rows, which may not cover all ATMs.
- Geographic location, holidays, or events were not included.
- Real-time ATM transactions were not considered.

Future Enhancements

- Introduce **Pink ATM Cards** for women with safety features.
- Design **Priority ATM Cards** for senior citizens with larger fonts and simplified steps.
- Incorporate IoT sensors for real-time ATM monitoring.

- Expand the dataset to include multiple cities, holidays, and ATM downtime for better accuracy.
- Integrate predictive maintenance for ATMs using advanced ML models.

REFERENCES / LINKS

Dataset

- GoMask.ai Platform – ATM Transaction Dataset: https://gomask.ai/marketplace/datasets/atm-transaction-usage-patterns?utm_source=chatgpt.com

Python & Libraries

- https://colab.research.google.com/drive/15PtzxLs-xKWU_YyNi67goSNA2DZaLyMm?usp=sharing

ATM Analytics / Refill Optimization Articles

- ATM Cash Refill Optimization using Data Analytics: https://dergipark.org.tr/en/pub/umagd/issue/39915/484670